

Commerce XI Maths App

Exercise 1.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 18

Question 1 - Enter answer · PG-15–16 · EX-Q1(i)

For data : 79, 82, 36, 38, 51, 72, 68, 70, 64, 63

Calculate D_6 .

Correct answer: 69.2

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 69.2.

Question 2 - Enter answer · PG-15–16 · EX-Q1(ii)

For data : 79, 82, 36, 38, 51, 72, 68, 70, 64, 63

Calculate P_{85} .

Correct answer: 80.05

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 80.05.

Question 3 - Enter answer · PG-15–16 · EX-Q2(i)

The daily wages (in ₹) of 15 labours are

230, 400, 350, 200, 250, 380, 210, 225, 375, 180, 375, 450, 300, 350, 250

Calculate D_8 .

Correct answer: 379

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 379.

Question 4 - Enter answer · PG-15–16 · EX-Q2(ii)

The daily wages (in ₹) of 15 labours are

230, 400, 350, 200, 250, 380, 210, 225, 375, 180, 375, 450, 300, 350, 250

Calculate P_{90} .

Correct answer: 420

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 420.

Question 5 - Enter answer · PG-15–16 · EX-Q3(i)

Calculate 2nd decile for the following :

x	80	100	120	145	200	280	310	380	400	410
f	15	18	25	27	40	25	19	16	8	7

Correct answer: 120

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 120.

Question 6 - Enter answer · PG-15–16 · EX-Q3(ii)

Calculate 65th percentile for the following :

x	80	100	120	145	200	280	310	380	400	410
f	15	18	25	27	40	25	19	16	8	7

Correct answer: 280

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 280.

Question 7 - Enter answer · PG-15–16 · EX-Q4(i)

From the following data calculate the rent of 15th house.

House Rent (in ₹)	11000	12000	13000	14000	15000	16000	17000	18000
No. of houses	25	17	13	14	15	8	6	2

Correct answer: 11000

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 11000.

Question 8 - Enter answer · PG-15–16 · EX-Q4(ii)

From the following data calculate the rent of 65th house.

House Rent (in ₹)	11000	12000	13000	14000	15000	16000	17000	18000
No. of houses	25	17	13	14	15	8	6	2

Correct answer: 14000

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 14000.

Question 9 - Enter answer · PG-15–16 · EX-Q4(iii)

From the following data calculate the rent of 91st house.

House Rent (in ₹)	11000	12000	13000	14000	15000	16000	17000	18000
No. of houses	25	17	13	14	15	8	6	2

Correct answer: 16000

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 16000.

Question 10 - Enter answer · PG-15–16 · EX-Q5(i)

The following frequency distribution shows the weight of students in a class.

Weight (in kg)	40	45	50	55	60	65
Number of Students	15	40	29	21	10	5

Find the percentage of students whose weight is more than 50 kg.

Correct answer: 30

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 30.

Question 11 - Enter answer · PG-15–16 · EX-Q5(ii)

The following frequency distribution shows the weight of students in a class.

Weight (in kg)	40	45	50	55	60	65
Number of Students	15	40	29	21	10	5

If the weight column provided is of mid values then,
find the percentage of students whose weight is more than 50 kg.

Correct answer: 42

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 42.

Question 12 - Enter answer · PG-15–16 · EX-Q6(i)

Calculate D_4 from the following data:

Mid Value	2.5	7.5	12.5	17.5	22.5	Total
Frequency	7	18	25	30	20	100

Correct answer: 13

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 13.

Question 13 - Enter answer · PG-15–16 · EX-Q6(ii)

Calculate P_{48} from the following data:

Mid Value	2.5	7.5	12.5	17.5	22.5	Total
Frequency	7	18	25	30	20	100

Correct answer: 14.6

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 14.6.

Question 14 - Enter answer · PG-15–16 · EX-Q7(i)

Calculate D_9 of the following distribution.

Length (in Inches)	0–20	20–40	40–60	60–80	80–100	100–120
No. of units	1	14	35	85	90	15

Correct answer: 98

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 98.

Question 15 - Enter answer · PG-15–16 · EX-Q7(ii)

Calculate P_{20} of the following distribution.

Length (in Inches)	0–20	20–40	40–60	60–80	80–100	100–120
No. of units	1	14	35	85	90	15

Correct answer: 58.86

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 58.86.

Question 16 - MCQ · PG-15–16 · EX-Q8

Weekly wages for group of 100 persons are given below :

Weekly wages (in Rs.)	0-500	500-1000	1000-1500	1500-2000	2000-2500
No. of persons	7	a	25	30	b

D_3 for this group is Rs. 1100. Calculate the missing frequencies a, b.

Options

- A. (18, 20)
- B. (20, 18)
- C. (17, 21)
- D. (22, 16)

Correct answer: (18, 20)

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is (18, 20).

Question 17 - MCQ · PG-15–16 · EX-Q9

The weekly profit (in rupees) of 100 shops are distributed as follows :

Profit per shop	0-1000	1000-2000	2000-3000	3000-4000	4000-5000	5000-6000	6000-7000
No. of Shops	10	16	26	20	20	5	3

Find the limits of the profit of middle 60% of the shops.

Options

- A. $(P_{20}, P_{80}) = (1625, 4400)$
- B. $(P_{20}, P_{80}) = (1500, 4500)$
- C. $(P_{30}, P_{70}) = (1625, 4400)$
- D. $(P_{20}, P_{80}) = (1600, 4600)$

Correct answer: $(P_{20}, P_{80}) = (1625, 4400)$

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is $(P_{20}, P_{80}) = (1625, 4400)$.

Question 18 - Enter answer · PG-15–16 · EX-Q10

In a particular factory, workers produce various types of output units.
The following distribution was obtained

Output units Produced	70–74	75–79	80–84	85–89	90–94	95–99	100–104
No. of workers	40	45	50	60	70	80	100

Find the percentage of workers who have produced less than 82 output units.

Correct answer: 24.72

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 24.72.